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Segment	Basics of Insulation
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Basics of Insulation

Heat always flows from higher to lower temperature an unavoidable physical law. In refrigeration and air conditioning, where spaces are maintained below ambient temperature, heat constantly tries to penetrate inward. Refrigeration reverses this natural phenomenon by pumping heat from a colder space to a warmer one, requiring external energy. **Cold insulation** is therefore essential wherever operating temperatures fall below ambient to prevent heat gain, condensation, and freezing.

What Is Thermal Insulation?

A material that reduces thermal conduction (heat ingress) and reflects thermal radiation, acting as a barrier between a cold interior and a warm exterior environment.

Temperature Range

Thermal insulation applies from **−75°C to +815°C**. Below −75°C is cryogenic; above +815°C is refractory. Between these extremes lie low, intermediate, and high temperature applications.

Why It Matters

Without insulation, refrigeration systems waste enormous energy fighting constant heat ingress. Insulation reduces energy consumption, prevents condensation, and protects equipment and personnel.

Heat Transfer & Temperature Ranges

Three Modes of Heat Transfer

- **Conduction**

  Transfer of heat molecule-to-molecule through a substance via chain collision. Most prominent in dense solids. Rate depends on thermal conductivity (W/m·K) and temperature difference. Higher density = higher conductivity (e.g., metals). Low-density materials with small air/gas voids have low conductivity ideal for insulation.

- **Convection**

  Heat transfer by movement of molecules from one place to another. Forced convection is driven by external influences such as wind, fans, and ventilators. Heat loss/gain to/from atmosphere occurs via convection.

- **Radiation**

  Heat passes from a source to an absorbent surface without heating the intervening space. Heat loss/gain from atmosphere also occurs via radiation. Reflective surfaces reduce radiant heat transfer.

Thermal Insulation Temperature Ranges

Category	**Range**	**Examples**
Low Temp 1	+15°C to +1°C	Chilled water, fresh fruit/vegetable cold rooms
Low Temp 2	0°C to −40°C	Refrigeration, glycol brine chillers, frozen food storage
Low Temp 3	−40°C to −75°C	Blast freezers, plate freezers, IQF, brine process plants
Cryogenic	−76°C to −273°C	Absolute zero range (not addressed here)
Intermediate 1	16°C to 100°C	Hot water, steam condensate
Intermediate 2	101°C to 315°C	Steam, high temperature water
High Temp	316°C to 815°C	Turbines, boilers, incinerators, exhausts, stacks

The R value defines insulation resistance. Lower thermal conductivity and higher thickness both increase R. Higher density increases heat capacity and lowers thermal diffusivity — slowing temperature rise when cooling is interrupted.

Why Insulation Is Necessary & Its Benefits

A thermal insulation system combines material, ancillaries, and application methodology to resist environmental heat flow into a space maintained below ambient temperature. The most effective insulation maximizes resistance (R value) while minimizing thermal conductivity. Moisture control is equally critical: as heat flows from high to low temperature, moisture travels from higher vapour pressure (outside) to lower vapour pressure (inside). Since moisture is a good thermal conductor, its presence in an insulation system is highly detrimental — causing increased energy costs, condensation, and often complete system failure. In most refrigeration applications, design thickness is dictated by moisture prevention rather than economic payback alone.

Energy & Temperature Control

- Reduces heat gain or heat loss
- Enhances process performance, reducing energy consumption
- Provides accurate control of process temperatures, minimising fluctuations
- Limits temperature change of process fluid
- Helps reduce carbon emissions

Moisture & Condensation

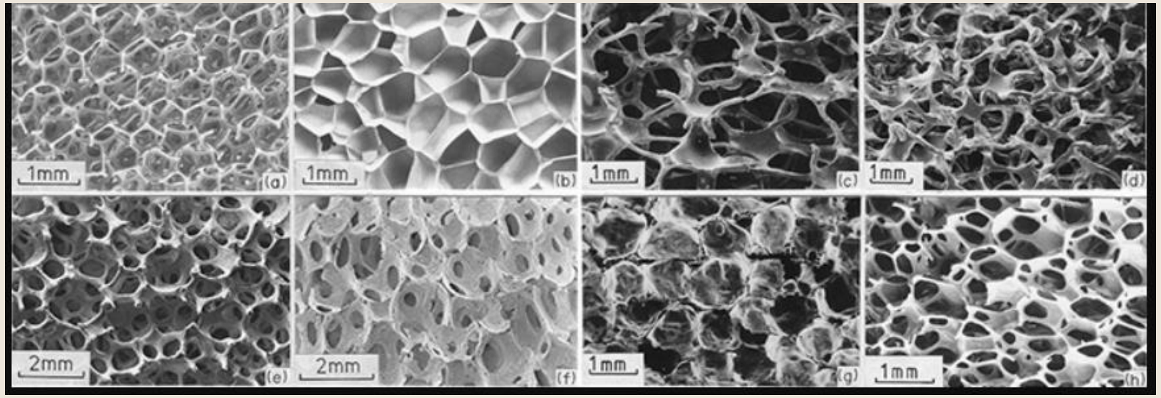
- Prevents moisture penetration; acts as a vapour barrier
- Prevents condensation and water dripping from ducting, piping, drain pans, ceilings
- Prevents gas condensation inside pipes
- Prevents ice accumulation and protects the freezer

Safety & Compliance

- Controls surface temperatures to avoid contact burns (hot or cold) — personnel protection
- Promotes fire safety by slowing the spread of fire in buildings
- Acoustical performance reduces noise levels in ducting and equipment
- Improves appearance of equipment and piping

Insulation Materials & Cell Structures

Insulation materials are classified by their cellular structure, which directly determines thermal performance. Closed cell foam has a higher R value than open cell foam and is generally the preferred choice for refrigeration applications.



Types of Insulation by Composition

❑ Fibrous

Small-diameter fibers (glass fiber, mineral wool, rock wool, slag wool, alumina silica) finely dividing air space. Fibers may be bonded with organic binders.

❑ Cellular

Small individual cells separated from each other. Materials: foam glass, polystyrene (closed cell), polyurethane, polyisocyanurate, elastomeric.

❑ Granular

Small nodules with voids; gas can transfer between spaces. Examples: calcium silicate, perlite, vermiculite, cellulose, diatomaceous earth, EPS.

❑ Vacuum

Used for cryogenic applications. Multiple layers (up to 60/inch) of aluminum, copper, or gold separated by small air spaces. Eliminates convection and virtually eliminates radiation.

❑ Reflective

Reflective paints and surfaces lower long-wave emittance, reducing radiant heat transfer from the surface.

Open Cell

Rockwool/mineral wool, glass wool, cork, melamine foam, EPS, PUF (indoor). Best for low-to-medium temperatures.

Semi-Open/Microporous

Polyisocyanurate (PIR), extruded polystyrene (XPS), polyethylene (PE). Intermediate performance.

Closed Cell

Foam glass, cross-linked polyethylene (XLPE), nitrile butadiene rubber foam (NBR), polyolefin foam, PUF (outdoor). Highest R value; preferred for refrigeration.

Flexible Elastomeric

Elastic polymers: NBR, EPDM, SBR, Neoprene. Deform under stress and return to original shape. Excellent for pipe insulation.

Cross-Linked Polyolefin/Polyethylene group: Polyethylene (PE), Polypropylene (PP), Polymethyl pentene (PMP), Polyethylene Vinyl Acetate (PVA).

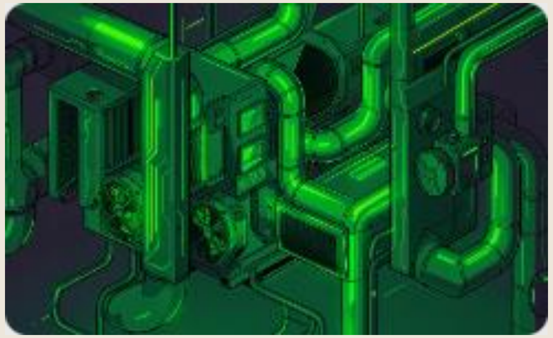
Insulation Applications in AC&R

Cold Storages & Freezers

The major use of insulation in cooling applications — walls, ceilings, flooring, and doors for positive and negative temperature cold rooms. Includes blast freezer rooms, IQF, and plate freezers.

Refrigerant Pipelines & Equipment

Refrigerant pipe lines below ambient temperature; shell and tube chillers (water/glycol/brine);inter-stage coolers; low pressure/temperature vessels; surge drums; knock-out drums; accumulators; suction/liquid line heat exchangers; pumps.



Air Conditioning Systems

Water chillers, refrigerant piping, air handling units (AHUs), and ducting. Package chillers installed on terraces use insulation for the casing.

Specialised Applications

Transport refrigeration truck bodies; ice plants (block ice, flake ice, tube ice makers); ice rinks and amusement park ice rooms; morgues; supermarket display cabinets.

Common Insulation Materials in the AC&R Industry

- Fiber Glass
- Mineral Wool
- Polystyrene
- Polyurethane
- Polyisocyanurate
- Polyolefin
- Elastomeric Rubber

Insulation forms include: rigid boards/blocks/pre-formed shapes (pipe insulation, curved segments); flexible sheets and pre-formed shapes; flexible blankets (fibrous); cements (hydraulic-setting or air-drying); and foams (poured, froth, or spray for irregular areas and flat surfaces).

Important Formulae for Insulation Material Selection

This chapter covers the essential formulae and properties used when selecting insulation materials for various applications. A key focus is the conversion of values between the SI and FPS systems critical because most heat load calculation software uses FPS units, while engineering practice increasingly demands SI. Understanding how these conversions are derived not just looked up in tables empowers engineers and designers to work confidently without relying on conversion charts.

SI vs. FPS Conversion

Transmission loads are often calculated in FPS and then converted to SI. This chapter explains the derivation behind each conversion factor.

Material Properties

Key thermal, vapour, and physical properties that determine the suitability of an insulation material for a given application.

Selection Criteria

Desirable properties of insulation materials including conductivity, resistance, fire rating, moisture resistance, and more.

Thermal Conductivity *k* and Thermal Resistance *R*

Thermal Conductivity *k* — W/(m·K)

Thermal conductivity (λ) is a specific material property representing heat flow in watts through 1 m² surface and 1 m thick flat layer of a material when the temperature difference between the two surfaces in the direction of heat flow is 1 Kelvin. In FPS units it is expressed per inch thickness; in SI it is per metre thickness — hence the unit is W/m·K, not W/m²·K.

Thermal conductivity increases with temperature and may depend on direction of heat transfer. ASTM standards give values at 24°C. To convert from FPS (Btu·in/h·ft²·°F) to SI:

$$k_{\mathrm{SI}} = k_{\mathrm{FPS}} \times 0.1442$$

Derived as: $\frac{1.055.056J(W.s) \times 3.2808ft \times 3.2808ft \times 1.8\ F}{3600 \times 3.2808ft \times 12in} = 0.1442$

Thermal Resistance *R* — m²·K/W

Thermal resistance describes the insulation effect of a constructional layer. It is calculated as:

$$R = \frac{d}{\lambda}$$

where *d* is thickness and λ is thermal conductivity (EN ISO 6946). For multi-layer building components, individual layer resistances are added together.

To convert from FPS (hr·°F·ft²/Btu) to SI (m²·K/W):

$$R_{SI} = R_{FPS} \times 0.17611$$

Derived as: $\frac{3600}{1055.056J\left(\frac{W}{s}\right) \times 1.8F \times 3.2808ft \times 3.2808ft} = 0.17611$

Overall thermal resistance of a structure must be calculated first before computing heat gain.

Thermal Conductance, Transmittance & Surface Film Conductance

Thermal Conductance C — W/m²·K

Thermal conductance is the **reciprocal of thermal resistance**. It equals conductivity multiplied by thickness. To convert from FPS to SI: multiply by **5.678** (= 1/0.17611). The key distinction: **conductivity** is a pure material property independent of dimensions, while **conductance** is an object property that depends on both material and thickness.

$$C(W/m^2.K) = C_{FPS} \times 5.678$$

Overall Heat Transfer Coefficient *U* — W/m²·K

Thermal transmittance is the heat flow in watts through 1 m² of a building component per 1 K temperature difference. *U = 1/R*. It represents the reciprocal of **all resistances** acting as barriers to heat travel from outside to inside the refrigerated space. To convert from FPS to SI:

$$U_{SI} = U_{FPS} \times 5.678$$

Surface Film Conductance *f* — W/m²·K

Every material surface offers resistance to heat flow. This resistance depends on **reflectivity, roughness, orientation** (horizontal or vertical), length, and air velocity over the surface. The reciprocal of this surface resistance is the **surface film conductance** (*f*), expressed in the same units as conductance (W/m²·K or Btu/hr·ft²·°F).

Heat Capacity, Heat Flux & Thermal Diffusivity

Volumetric Heat Capacity *C_p* — J/m³·K

The change in heat stored in a unit volume (m³) of material for a unit change in temperature (K).

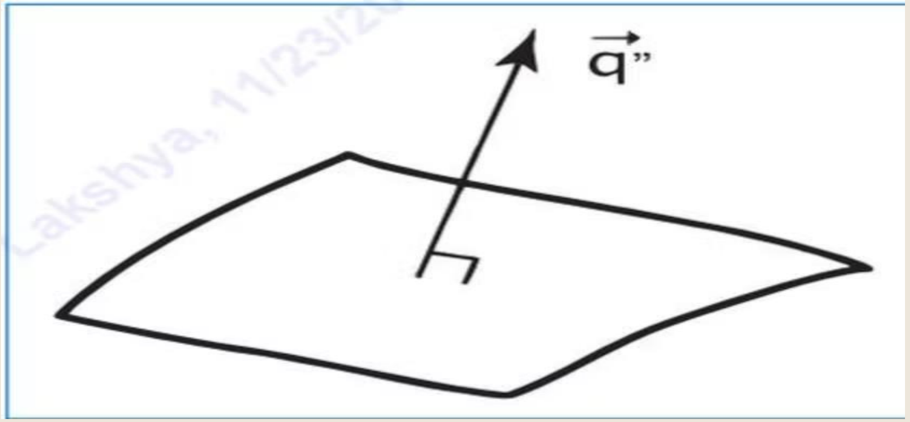
Specific Heat Capacity *c* — J/kg·K

The amount of heat needed to raise 1 kg of material by 1 K. A **good insulator has high specific heat capacity** — it takes longer to absorb heat before its temperature rises. This is the basis of thermal mass and thermal buffering (decrement delay).

Material	c = (J/kg·K)
Water	4190
Wood and wood-based materials	1600
Rigid polyurethane foam (PUR/PIR)	1400–1500
Wood-fibre insulation boards	1400
Mineral wool	1030
Air (ρ = 1.25 kg/m^3)	1000
Aluminium	1000
Other metals	800

Heat Flux *q"* — W/m²

Also called heat flux density or heat flow rate intensity — it is the **flow of energy per unit area per unit time**. In SI, its unit is watts per square metre (W/m²).



Thermal Diffusivity $\propto$= mm²/s

While thermal conductivity measures a material's ability to *conduct* heat, thermal diffusivity measures its **thermal inertia** — the ability to conduct thermal energy *relative to* its ability to store it. Insulators should have **low thermal diffusivity**.

$$\propto = \frac{Thermal\ conductivity}{density\ \times specific\ heat\ capacity}$$

Copper: 98.8 mm²/s (fast transmitter, cold to touch). Wood: 0.082 mm²/s (slow transmitter, good insulator). The lower the value, the better the insulator.

Thermal Emissivity & Vapour Resistance

Thermal Emissivity, ε

Emissivity is the fraction of infrared energy emitted by a surface relative to a perfect black body (emissivity = 1). A perfect thermal mirror has emissivity = 0. Most real materials fall between 0 and 1. For example, an object emitting 90 out of 100 possible units has emissivity = 0.9.

Placing a **low-emissivity material** (aluminium or copper) in insulation acts as a **radiant barrier**, reducing infrared radiant heat reaching the insulation below. This can reduce total heat flux through a ceiling by up to **50%** — especially effective in warm climates. Emissivity varies with surface texture, colour, and temperature.

Reflective thermal insulation is typically aluminium foil with core materials such as low-density polyethylene foam or fibreglass. Aluminium foil facing can stop **97% of radiant heat transfer**.

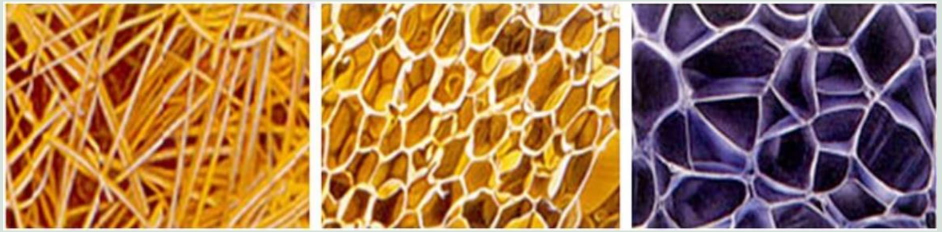
Vapour Resistance & μValue

Vapour resistance measures a material's reluctance to let water vapour pass through, accounting for thickness. Measured in **MN·s/g**. Vapour resistivity is measured in MN·s/g·m.

The μ value (water vapour resistance factor) is dimensionless — measured relative to still air. To calculate:

1. Multiply vapour resistance by 0.2 gm/MN·s (UK vapour permeability of still air)
2. Divide by thickness in metres

Example: Vapour resistance = 10,000 MN·s/g, thickness = 100 mm:
μ = *10,000 MN.s/g x 0.2gm/MN.s ÷0.1 = 20,000* (Ref: BS 5250:2002 Annex E)



Open cell (low μ), semi-closed cell (moderately high μ), closed cell (high μ above 8000).

Permeability, Permeance & Moisture Transmission

Permeability — kg/Pa·s·m

Describes moisture transmission of a material. It is a **theoretical number** calculated by multiplying Permeance by the material's thickness (stated in perm inches). Permeability is **not dependent on material thickness.** Example: aluminium foil has a permeability of 0.05 US perm = 2.9 SI perm.

Permeance (Perm)

The moisture transmission rate of a membrane — a **performance measure** expressed in perms. Applies to any material with water vapour transmission rate less than 1 perm. Dividing permeability by thickness gives permeance. **Lower numbers = better moisture resistance.**

1.0 US perm = 1.0 grain/ft^2/hr/in Hg (7,000 grains = 1 pound)

1.0 perm = 57 SI perm = 57 ng/s·m^2·Pa

Permeance to Water Vapour (*pv*)

Quantity of water vapour passing through unit area of unit thickness when the difference in water pressure between both faces is one unit. Expressed as g·cm·mmHg^{-1}·m^{-2}·day^{-1} or in SI as g·m·MN^{-1}·s^{-1}.

Resistance to Water Vapour (*rv*)

The reciprocal of permeance to water vapour:

$$r_v = {}^{1}/_{p_v}$$

Higher resistance means the material is more effective at blocking moisture transmission.

Which Material is Suitable as Insulation?

Closed-cell insulation is the most commonly specified material for cold work it offers resistance to water vapour and often has better thermal conductivity (k value) than fibrous alternatives. Materials with low thermal conductivity have a **high proportion of small voids containing air or gas** too small to transmit heat by convection or radiation.

It is very important to understand that insulation material itself has hardly any insulating property. It is the **trapped air/gas bubbles inside the insulation** that impart it the insulating properties. The more the trapped air bubbles, the better the insulation.

Density & Void Relationship

If density is **low**, air/gas voids are comparatively large — best insulation for low to medium temperatures where compression and vibration are not critical factors. If density is increased **too high**, the solid content overcomes the insulating effect of the voids the material actually becomes a conducting medium, increasing heat flow and ingress.

Natural vs. Man-Made

Thermal insulation materials are generally man-made, but natural substances are used in certain applications. The key principle remains the same: maximise trapped air or gas within the structure to minimise heat transfer. Insulation materials must be installed **dry** and kept dry when wet, air enclosures fill with water (a better conductor), increasing conductivity significantly.

Desirable Properties of Insulation Material

Low Thermal Conductivity

Light materials with dry air enclosures are better insulators. Conductivity increases with moisture keep insulation dry during and after installation.

High R-Value & Water Resistance

Must trap air inside its structure. Should have very low moisture permeability and water retention minimizing condensation and corrosion.

Non-Flammable & Fire Resistant

British Standard 476 classifies materials from A1/A2 (non-combustible) to F (easily flammable). **FM (Factory Mutual) approval** for fire resistance is a must for commercial applications.

Physical & Environmental Properties

Wind and UV resistant; light in colour (low emissivity, reflects solar radiation); sufficient compressibility, strength, and load-bearing capacity; low density and lightweight; durable against pests and rodents; easy to apply; locally available. Stringent insulation can cut CO_2 emissions by 5% (Kyoto requirements).

Summary: Key Characteristics for Selection

Symbol	Factor	Unit	Goal
k	Conductivity	W/m·K	Lower
C	Conductance	W/m²·K	Lower
U	Overall heat transfer coeff.	W/m²·K	Lower
D	Thermal diffusivity	m²/s	Lower
E	Permeability	Perm	Lower
ε	Thermal emissivity	Dimensionless	Lower
f	Surface film conductance	W/m²·K	Lower
ρCp	Volumetric heat capacity	J/m³·K	Lower
q"	Heat flux	W/m²	Lower
R	Resistance	m²K/W	Higher
μ	Vapour resistance	MN·s/g·m	Higher
c	Specific heat capacity	J/kg·K	Higher

Properties of Insulation Materials

A comprehensive technical guide to natural and synthetic insulation materials covering thermal properties, acoustic performance, fire safety, applications, and handling for residential, commercial, and industrial use.

01

Natural Materials

Fibreglass, Glass Wool, Mineral Wool, Cellulose, Aerogel

02

Synthetic Materials

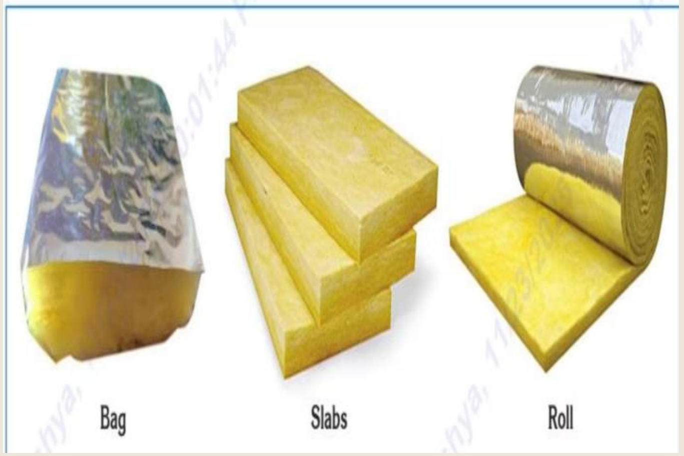
Polystyrene (EPS & XPS), Polyurethane (PUF), Polyisocyanurate (PIR)

03

Advanced Systems

Sandwich Panels, Flexible Closed-Cell Foams, Cross-linked Polyolefin, Vacuum Insulation

Fibreglass



Key Properties

- Thermal conductivity λ = **0.035 W/m·K**
- Thermal resistance at 100 mm = **2.85 K·m²/W**
- Specific heat capacity = **1,030 J/(kg·K)**
- Density = **12 to 80 kg/m³**
- Thermal diffusivity = **1.6×10^{-6} m²/s**
- Embodied energy = **26 MJ/kg**
- Vapour permeable: **Yes**

Manufacturing & Forms

Made from molten glass and silica rocks spun into fibres, with 20–30% recycled industrial waste. Loose-fill variants use 40–60% recycled glass. The process traps air pockets between fibres, producing high thermal resistance. Sold as **blankets (rolls/batts)**, **slabs**, and **loose fill**, often with paper or aluminium foil backing. Also available as pipe sections for chilled water piping.

Performance & Safety

Conserves 12× more energy than is lost in production; may reduce residential energy costs by up to **40%**. Non-combustible. Per IARC/WHO (Joint Monograph, Vol. 81, 2002), glass wool is **Class 3 — not classifiable as carcinogenic**. Installers must wear eye protection, mask, and gloves. Some formulations still use formaldehyde binder, which may be carcinogenic.

Advantages

- Inexpensive, effective, and does not shrink
- Non-combustible; insects do not consume it
- Available in low, medium, and high density (R-11 to R-15 for 2× 4 wall)
- Sealed plastic-film batts act as effective vapour barrier
- Some variants use recycled glass, reducing ecological footprint
- Blown fibreglass provides consistent coverage in wall cavities

Disadvantages

- Protective gear essential — slivers can lodge in skin or be inhaled, irritating alveoli
- Requires vapour barrier unless plastic-sealed batts are used
- Blankets do not seal wall/ceiling spaces very tightly
- Low-density fibreglass settles and sags, reducing R-value over time
- Cramming into smaller spaces requires venting to avoid moisture build-up

NATURAL INSULATION

Fibreglass Duct Boards & Glass Wool

Fibreglass for Air Ducts

Fibreglass is applied over GI sheet ducts using bands and stick pins, typically as aluminium foil laminated rolls and slabs. Fibrous glass insulation serves four key functions in air duct systems:

Temperature Control

Delivery of cooled air at comfort levels suited to building occupancy

Acoustical Control

Absorbs noise from central air handling equipment and cross-talk between spaces

Condensation Control

Prevents condensation on ducts when installed R-value recommendations are followed

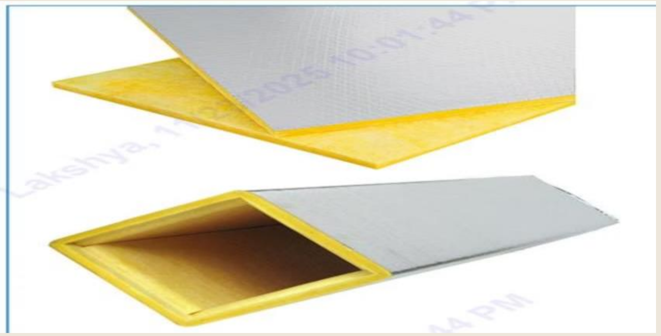
Energy Conservation

Reduces HVAC operating costs by controlling heat loss/gain through duct walls

Fibreglass duct boards (48 kg/m³ density) with glass tissue and glass-reinforced aluminium foil laminations are used for prefabricated ducting in buildings and marine applications, and for acoustic lining in sound recording studios and TV broadcasting facilities. Properties: λ = **0.041 W/m·K**; density = **48 kg/m³**; specific heat = **1,000 J/(kg·K)**; vapour permeable: **No**. Suitable from −100°C to +230°C; up to 450°C with high-temperature binder.

Glass Wool

Glass wool is manufactured from fused borosilicate glass directed to a spinning machine, forming long spun fibres applied with chemical binder, uniformly distributed and cured to produce blankets or boards. The laminar structure with interconnected air pockets resists air flow, reducing sound and heat transmission. 100% of raw material is converted to insulation with no impurities.



Thermal Conductivity by Density

Density (kg/m^3)	λ (W/m·K) at 25°C	R (m^2·K/W) at 25 mm
12	0.041–0.045	0.55–0.61
24	0.033–0.036	0.69–0.76
48	0.030–0.032	0.78–0.83

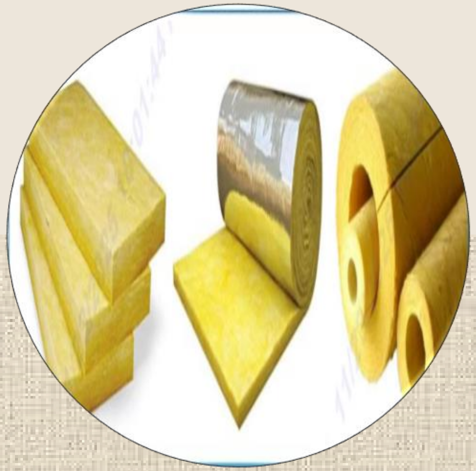
Glass wool is **non-combustible** (manufactured at 1,100°C), rated **Class O** per BS 476 Parts 6 & 7, and **A1 reaction to fire** per EN 13501-1. It is chemically inert (pH ≈ 7), free from sulphur and chloride, moisture absorption <2%, and mould/vermin resistant. Noise Reduction Coefficient (NRC) reaches **0.95** at 12 kg/m³ / 85 mm thickness.

NATURAL INSULATION

Mineral Wool, Rockwool, Cellulose & Aerogel`

Mineral Wool

Mineral wool encompasses glass wool (from recycled glass), rock wool (from basalt), and slag wool (from steel mill slag). R-value: **2.8 to 3.5**. Properties: λ = **0.032–0.044 W/m·K**; density up to **120 kg/m³**; temperature range **200–800°C**; vapour permeable. Non-combustible; classified **Group 3** (not carcinogenic) per IARC/WHO. A NAIMA study (1997) and Duke University medical study (2001) confirmed glass wool does not affect indoor air quality.

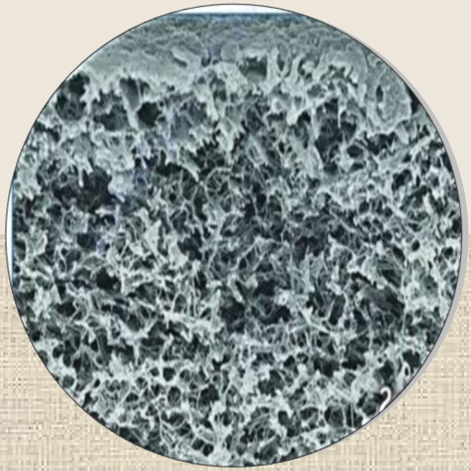


Rockwool

Made from melted basalt rocks (oxides of iron, silica, alumina) bonded with phenol formaldehyde resin at 1,600°C, spun into chemically inert fibres. Properties: λ = **0.029–0.044 W/m·K**; density **40–160 kg/m³**; embodied energy **3.1 MJ/kg**; temperature range **—100 to +800°C**; vapour permeable. R-values range from **1.72 to 4.14 m²·°C/W**. Fire rated for **one hour** at 80–100 mm / 160 kg/m³; **A-60 classification**. Water repellent per BS 2972. Non-carcinogenic (Group 3). Available as slabs, rolls, blankets, boards, pipe sections, and loose wool.

Cellulose

One of the most eco-friendly insulation forms — a plant-based material and the oldest form of home insulation, now produced from recycled newspapers treated with fire-retardant chemicals. R-value: **3.1 to 3.7**. Properties: λ = **0.035 W/m·K** (lofts), **0.038–0.040 W/m·K** (walls); specific heat = **2,020 J/(kg·K)**; density **27–65 kg/m³**; embodied energy = **0.45 MJ/kg** — the lowest of any insulation. Highest recycled content of any insulation material. Studies by the National Research Council Canada confirm cellulose may protect buildings from fire better than fibreglass due to its density restricting oxygen. Chemical fire-retardant additives may irritate skin; breathing mask required during handling.

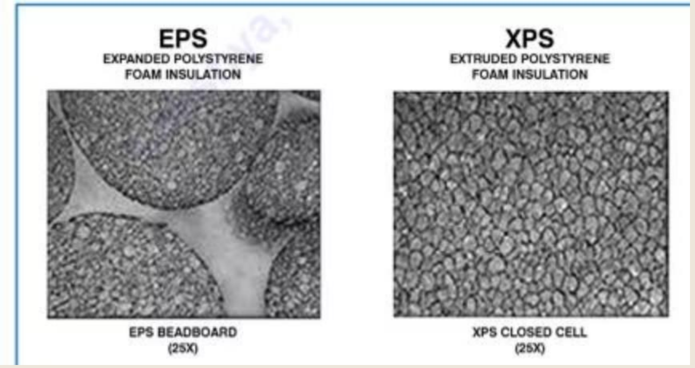


Aerogel

A synthetic ultra-light porous material (nicknamed "frozen smoke") derived from a gel where liquid is replaced with gas. Silica aerogel — the most common type — has a solid matrix comprising only **3% of volume**, with 97% air in nano-pores. Properties: λ = **0.014 W/m·K** (lowest of all listed materials); thermal resistance at 50 mm = **3.8 K·m²/W**; density = **150 kg/m³**; embodied energy = **53.9 MJ/kg**; vapour permeable. **Cryogel** is used for sub-ambient/cryogenic applications (down to −273°C); **Pyrogel** for high-temperature applications (up to +650°C).

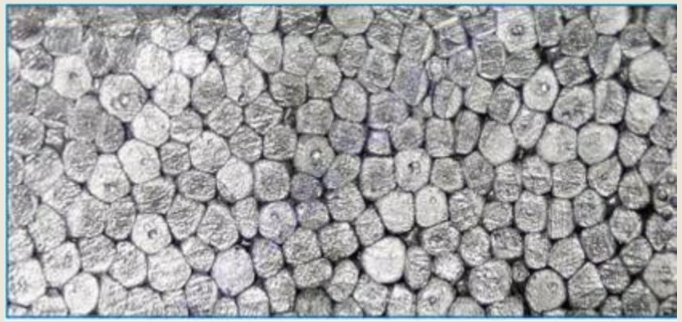
SYNTHETIC INSULATION

Polystyrene: EPS & XPS



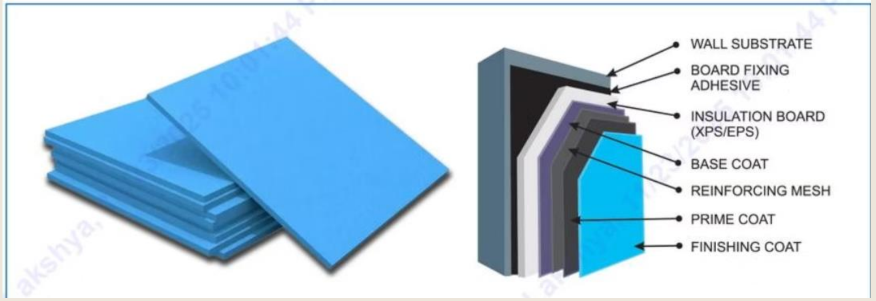
Expanded Polystyrene (EPS)

EPS is a rigid, closed-cell thermoplastic foam produced from solid polystyrene beads. Small amounts of gas within the beads expand when steam is applied, forming closed cells occupying ~**40× the original bead volume**. Pre-formed beads can be converted to large blocks, custom-fabricated shapes, or pipe sections. EPS beads mixed with cement produce lightweight insulating mortar for hollow wall cavities. Properties: density **21 kg/m³**; λ (max) = **0.0376 W/m·K**; compressive strength **102 kPa**; water absorption (max) **1.13% v/v**; perm rating **5**. Made of **98% air**, making it one of the lightest materials available.



Extruded Polystyrene (XPS)

XPS begins with crystal polystyrene granules combined with additives and a blowing agent, melted under high temperature and pressure, then forced through an extruder die to expand into foam, which is shaped, cooled, and trimmed. Properties: density **34 kg/m³**; λ (max) = **0.0289 W/m·K**; compressive strength **350 kPa**; water absorption (max) **0.3% v/v**; perm rating **1**; flammability Class **B2** (DIN 4102). XPS perm rating drops from 1.1 → 0.7 → 0.6 as thickness increases from 25 → 50 → 75 mm.



EPS vs XPS: When to Choose

XPS is preferred when very good insulation, high compressive strength, negligible water absorption, and flammability performance are critical. **EPS is preferred** for lower density, lower cost, and easy availability with good insulation properties. Both are HFC, CFC, and HCFC free; pentane blowing agent has GWP <5. Both are 100% recyclable and non-toxic. Standard board sizes: 1,250×600 mm to 2,400×1,200 mm; thickness 20–100 mm.

SYNTHETIC INSULATION

Polyurethane Foam (PUF) & Polyisocyanurate (PIR)

Polyurethane Foam (PUF)

PUF is formed by a chemical reaction using a non-CFC blowing agent, conforming to IS 12436. It contains a low-conductivity gas in its cells and is relatively light (~2 lb/ft³). Available in **closed-cell** and **open-cell** formulations.

- **Closed-cell PUF**: >90% closed cells; density 32–60 kg/m³; λ = **0.017–0.023 W/m·K**; thermal resistance at 100 mm = **4.50 K·m²/W**; negligible moisture absorption; used outdoors and in high-humidity environments. R-value ~6.3 per inch.
- **Open-cell PUF**: density 7–14 kg/m³; λ = 0.034–0.039 W/m·K; vapour-permeable; intended for indoor walls, roofs, and acoustic comfort. R-value ~3.6 per inch.

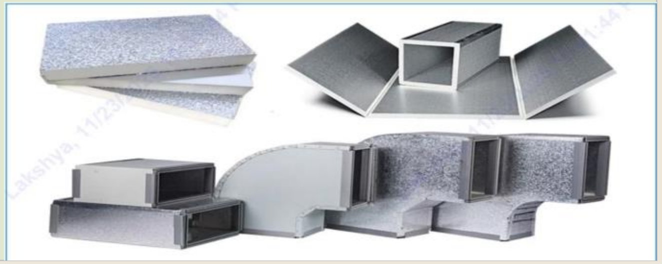
PUF is self-extinguishing, Class P rated (BS 476 Parts 6 & 7), embodied energy = **101 MJ/kg**, vapour permeable: **No**. Diffusivity: 0.0018–0.0024. Available as slabs, pipe sections, and spray. R-values: **1.43–3.57 m²·°C/W**.

Polyisocyanurate (PIR)

PIR is manufactured by the same process as PUF but further heated with chemicals to enhance fire safety. It has a higher proportion of MDI and uses a polyester-derived polyol. PIR index is typically >180 (vs ~100 for PUR). Properties: λ = **0.019–0.023 W/m·K**; thermal resistance at 100 mm = **4.50 K·m²/W**; density 32–60 kg/m³; closed cell content ≥95%; water absorption **0.03%**; air velocity up to 35 m/s; coefficient of friction 0.0135.

Fire performance: **Class O** per BS 476 Parts 6 & 7; surface spread of flame **Class 1** per BS 476 Part 7; smoke development **Class A** per ASTM E-84; toxicity per NES: **1.0142**; smoke per NES: **780.89**. CFC and HCFC free, zero ODP. PIR panels of 100 mm thickness achieve **one-hour fire rating** per BS EN 1364-1:2015.

Over time, PIR R-value can drop as low-conductivity gas escapes (thermal drift/ageing), mostly within the first two years. Foil and plastic facings stabilise R-value; metal foil-faced panels show no change after 10 years. Reflective foil facing an open-air space can add an additional **R-2** to overall thermal resistance.



SYNTHETIC INSULATION

PUF/PIR Sandwich Panels & Spray Foam

PUF Spray Application

Spray-applied rigid PUF combines highly efficient thermal insulation with ease of application. It adheres strongly to almost any surface, does not sag or mat, and retains insulation value for the life of the installation. The seamless, monolithic nature seals cracks and renders surfaces moisture-resistant and draught-proof. Applied using two-component spray machines maintaining mix ratio to ±2% accuracy. Conforms to IS 13205 and IS 12432.

Properties: density **40–50 kg/m³**; compressive strength >205 kPa; thermal conductivity **0.023 W/m·K** at 10°C; closed cell content **92%**; water absorption (96h) max 2%; temperature limit 100°C. R-value at 75 mm = **3.57 m²·°C/W**. A UV-protective coating is required for exterior applications.

Sandwich Panels

Double-skin insulated sandwich panels are manufactured with rigid PUF or PIR sandwiched between pre-painted galvanised steel skins. The chemical blend (polyol + MDI + blowing agent) is injected under pressure and solidifies irreversibly. Panels are produced on **continuous** or **discontinuous (batch)** lines.

Thermal and load characteristics:

Thickness (mm)	60	80	100	150	200
U-value ($W/m^2 \cdot K$)	0.36	0.26	0.21	0.14	0.11
Panel wt. (kg/m^2)	11.25	12.05	12.85	14.85	16.85

Cam lock panels use male/female cam lock fasteners embedded at panel edges for tongue-and-groove assembly, providing rigidity, weatherproofing, and ease of installation. Preferred for small cold rooms and blast freezers where high air pressures are present. **Continuous panels** (12–20 m long) use a snap-lock or similar mechanism and are produced fully automatically with built-in cooling, inspection, and strap packing.

SYNTHETIC INSULATION

Flexible Closed-Cell Foams, Cross-linked Polyolefin & Vacuum Insulation

Flexible Closed-Cell Insulation for HVAC & Refrigeration

Used on chilled water piping, ducting, refrigerant piping, heat exchangers, and vessels. The key requirement is preventing condensation on cold surfaces, which can damage building materials and cause mould growth. Flexible closed-cell structure provides superior water vapour diffusion resistance of **μ ≥ 7,000** (μ ≥ 10,000 with anti-microbial protection). Fully vapour-sealed using contact adhesives — adhered seams form no thermal or vapour bridges.

Technical data: service temperature **−50°C to +110°C**; thermal conductivity at 0°C = **0.033 W/(m·K)**; fire performance **Class O** (BS 476 Part 6) and **Class 1** (BS 476 Part 7); FM Approved; zero ODP and GWP; dust and fibre free; in-built antimicrobial protection.

Four critical requirements for cold-line insulation: effective water vapour barrier, non-wicking cell structure, no thermal bridging at valves/flanges, and long-term stability of thermal values.

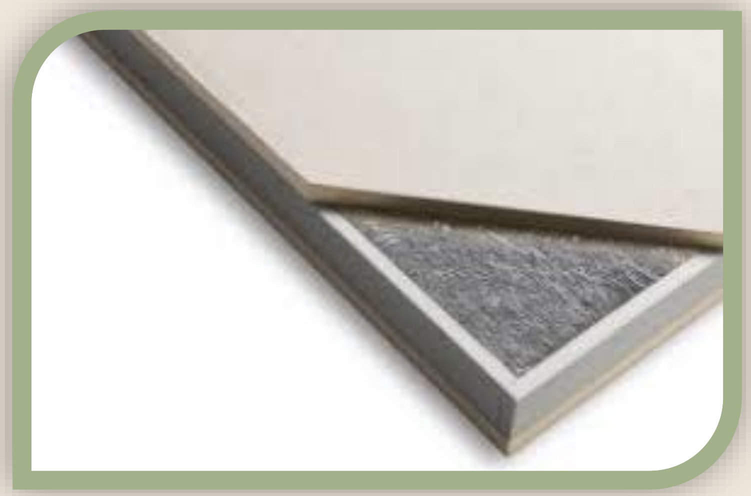
Cross-linked Polyolefin Foam

Physically cross-linked polyolefin foam uses electron beam bombardment to cross-link molecules, producing very fine, uniform closed cells. This controls thermal conductivity, water absorption, and vapour permeability. Properties: density **25 kg/m³**; λ = **0.028 W/m·K** at 0°C; fire rating **Class O** (BS 476 Parts 6 & 7); FM 4924 Approved; NFPA 90A/90B compliant; water absorption <0.8%. Temperature range: **−80 to +100°C**. CFC/HCFC free; complies with RoHS and REACH directives.

Chemically cross-linked PE foam uses chemical induction for cross-linking. Minimum density is limited to **30 kg/m³**, giving a higher k-value of 0.037 W/m·K at 23°C. Water vapour permeability resistance and water absorption are inferior to the physically cross-linked variant.

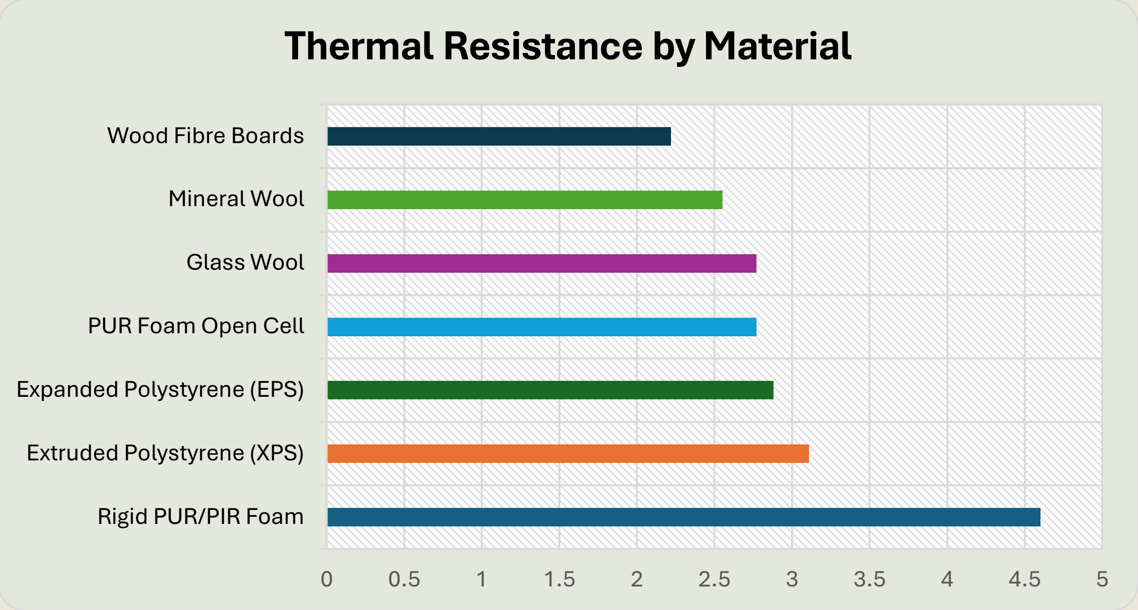
Vacuum Insulation Panels (VIPs)

VIPs consist of a core filler encapsulated by a super-barrier metal foil film, evacuated to **0.13–1.30 Pa** and sealed. The vacuum eliminates molecular movement needed for heat transfer. The core withstands external atmospheric pressures up to **101.3 kPa**. Water and gas adsorbing materials inside the panel maintain the vacuum for the intended service life. VIPs significantly outperform all closed-cell foams, foam beads, and fibre blankets in thermal resistance per unit thickness.



Thermal Performance Comparison

The chart below compares the thermal resistance (R-value at 10 cm thickness) of commercially available insulation materials. Rigid polyurethane/polyisocyanurate foam leads all materials by a significant margin.



Key Takeaways

- PUR/PIR foam offers the highest thermal resistance of all conventional materials at equivalent thickness
- Aerogel (λ = 0.014 W/m·K) outperforms all listed materials but carries the highest embodied energy (53.9 MJ/kg)
- Cellulose has the lowest embodied energy (0.45 MJ/kg) and highest recycled content
- Rockwool and glass wool offer the widest operating temperature ranges and best fire performance among natural materials
- Vacuum Insulation Panels (VIPs) represent the next generation, far exceeding all conventional systems

Thermal Conductivity Comparison (mW/m·K)

Material	λ (mW/m·K)
PU Spray Foam	24
XPS	35
EPS	40
Mineral Wool	40
Cork	45

Summary of Insulation Material Properties

The table below summarizes the key physical properties of the principal insulation materials covered in the 3rd chapter, enabling direct comparison for specification and selection purposes.

Property	PIR	PUF	EPS	Rockwool	Glass Wool	Aerogel
IS Code	IS 12436	IS 12436	IS 4671	IS 8183	IS 8183	—
Density (kg/m^3)	30–38	34–38	15–35	Slab: 48	Slab: 32	150
λ at 10°C (W/m·K)	0.023	0.023	0.037	0.033	0.033	0.014
Water absorption (%)	0.1	0.1	1	2.3	2.3	—
Compressive strength (kN/m^2)	115	115	100	—	—	—
Closed cell content (%)	≥86	≥85	—	—	—	—
Horizontal burn (mm max)	25	125	—	Non-comb.	Non-comb.	—
Vapour permeable	No	No	No	Yes	Yes	Yes
Embodied energy (MJ/kg)	101	101	19	3.1	26	53.9

Best Thermal Performance

PIR/PUF (λ = 0.023 W/m·K) and Aerogel (λ = 0.014 W/m·K) lead all materials in thermal efficiency

Best Fire Safety

Rockwool and Glass Wool are non-combustible; PIR achieves Class O and one-hour fire rating at 100 mm

Best Eco-credentials

Cellulose (0.45 MJ/kg embodied energy, highest recycled content); Rockwool (3.1 MJ/kg, recyclable)

Best Moisture Resistance

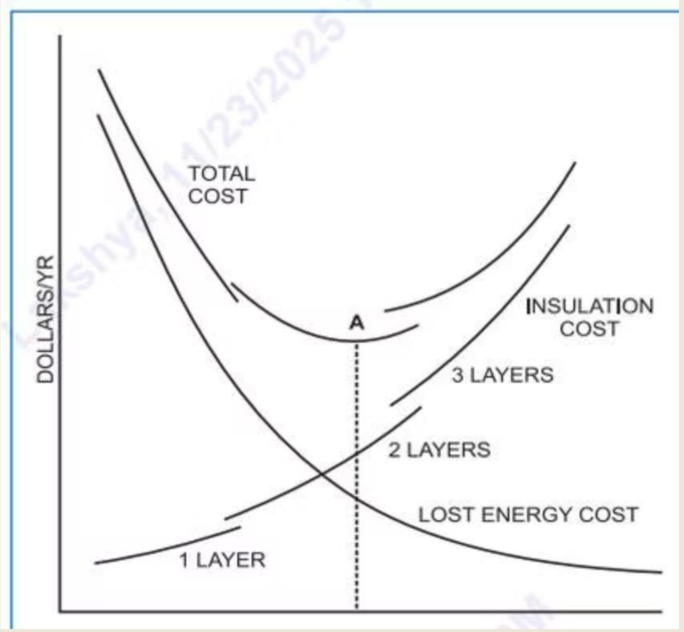
Closed-cell PUF/PIR and XPS with water absorption <0.1–0.3%; flexible closed-cell foams with μ ≥ 10,000

Insulation Thickness

This section presents recommended insulation thicknesses for different insulation materials used in cold rooms and piping, based on outdoor and indoor temperatures. Two primary objectives guide the selection of insulation thickness:

Optimised Thickness

Reduce heat gain/loss to the minimum practical level. The insulation thickness that achieves the minimum total cost balancing insulation cost against lost energy cost is taken as the optimum insulation thickness.



Moisture Control

Prevent moisture transmission and condensation on cold surfaces. Since moisture is a good thermal conductor, its presence in an insulation system is highly detrimental and can lead to complete system failure.

Optimum insulation thickness — point A marks the minimum of the total cost curve, balancing insulation cost (rising) against lost energy cost (falling).

Transmission Load & Heat Gain Fundamentals

Heat transfer through conduction directly depends upon thermal conductivity, surface area, temperature differential, and time. It varies inversely with the thickness of the material the greater the thickness, the less the heat ingress. Heat transmission into a refrigerated space through its ceiling, floor, and walls depends on:

(a) outside surface area of the cold room.

(b) temperature differential between the cold room and its surroundings.

(c) wall construction details and thermal conductivity of the insulation used.

Heat Gain Equation

$$Q = U \times A \times \Delta T$$

Q = Overall heat gain (W or BTU/h)

U = Overall heat transfer coefficient ($W/m^2 \cdot K$ or $BTU/h \cdot ft^2 \cdot °F$)

A = External area of envelope — four walls, ceiling and floor (m^2 or ft^2)

ΔT = Difference between outside air temperature and refrigerated space temperature (°C or °F)

Overall U-Value Calculation

To calculate the U-value, add all resistances acting as barriers to heat flow, then take the reciprocal:

$$R_{total} = 1/F_i + x_1/k_1 + x_2/k_2 + x_3/k_3 + 1/F_o$$

$$U = 1 / R_{total}$$

Where $\mathbf{x_1, x_2, x_3}$ = thicknesses of wall layer materials; $\mathbf{k_1, k_2, k_3}$ = thermal conductivities; $\mathbf{F_i}$ = inner surface convective coefficient; $\mathbf{F_o}$ = outer surface convective coefficient.

Worked Example: Calculating Total Resistance & U-Value

Consider a cold storage constructed of brick walls with expanded polystyrene (EPS) insulation, finished with sand and cement plaster. R-values are taken from ASHRAE Fundamentals 2013, Chapter 26. Polyurethane board (R-11 expanded) has a thermal conductivity of 0.16–0.18 BTU·in/h·ft²·°F (0.023–0.026 W/m·K).

Item	Thickness (in)	R ($ft^2 \cdot h \cdot °F/BTU$)	Thickness (m)	R ($m^2 \cdot K/W$)
F_o — 15 mph wind	—	0.166	—	0.0909
Cement plaster (outside)	½"	0.1	20 mm	0.017
Brick wall (100 lb/ft^3)	18"	3.6	450 mm	0.489
4" EPS Insulation	4"	**22.22**	100 mm	**4.347**
Cement plaster (inside)	½"	0.1	20 mm	0.017
F_i — still air	—	0.625	—	0.105
Total Resistance	—	**26.811**	—	**5.066**

The resistance of the insulating material (shown in bold) is far greater than all other resistances combined. It is therefore common practice to ignore the resistances of surface films, since their overall effect is very small (Source: ASHRAE Refrigeration 2010, Chapter 24.1). With modern PUF panels ≥100 mm, only the insulation resistance need be considered. To convert: $ft^2 \cdot h \cdot °F/BTU \times 0.17611 = m^2 \cdot K/W$.

Recommended Insulation Thicknesses for Cold Storages

The most important consideration is achieving the expected overall heat transfer coefficient for economic insulation thickness, irrespective of the insulation material. Reference Standard IS 661: 2000 gives recommended thicknesses for various insulation materials at a design ambient temperature of **+40°C and 50% RH**. The chapter covers rigid PUF/PIR panels for flat surfaces and closed-cell rigid/flexible insulation for round pipe surfaces.

Table 4-2: Recommended R and U Values per IS 661: 2000

Storage Temp (°C)	Exp. Wall R	Exp. Wall U	Int. Wall R	Int. Wall U	Roof R	Roof U	Floor R	Floor U
-30 to -20	5.88	0.17	2.12	0.47	7.14	0.14	5	0.2
-20 to -15	4.76	0.21	2.12	0.47	5.88	0.17	4.34	0.23
-15 to -4	4.34	0.23	2.12	0.47	4.76	0.21	3.7	0.27
-4 to +2	3.70	0.27	1.72	0.58	4.16	0.24	3.44	0.29
+2 to +10	2.85	0.35	1.07	0.93	3.44	0.29	2.12	0.47
+10 to +16	2.12	0.47	1.07	0.93	3.44	0.29	1.56	0.64
+16 and above	0.78	1.28	0.68	1.47	0.95	1.05	0.61	1.63

Insulation Thickness Tables: Walls, Roofs & Floors

Tables 4-3 through 4-6 give the required insulation thickness (mm) for exposed walls, intermediate walls, roofs, and floors at various storage temperatures. All values are per IS 661: 2000 at +40°C ambient and 50% RH. **Minimum available thickness: 25 mm for most materials (in 5 mm increments); PUF/PIR panels minimum 50 mm.*

Table 4-3: Exposed Walls (mm)

Temp (°C)	PUF/PIR	Phenolic	EPS	Rock wool	Glass wool	PUF Panel
-30 to -20	120	140	200	180	180	130
-20 to -15	100	110	160	150	150	100
-15 to -4	90	100	150	130	130	90
-4 to +2	80	90	120	110	110	80
+2 to +10	60	60	90	80	80	60
+10 to +16	40	50	70	60	60	50

Table 4-4: Intermediate Walls (mm)

Temp (°C)	PUF/PIR	Phenolic	EPS	Rock wool	Glass wool	PUF Panel
-30 to -20	50	50	70	60	60	50
-20 to -15	50	50	70	60	60	50
-15 to -4	50	50	70	60	60	50
-4 to +2	40	40	60	50	50	40
+2 to +10	20	20	30	30	30	20*
+10 to +16	20	20	30	30	30	20*

Table 4-5: Roofs (mm)

Temp (°C)	PUF/PIR	Phenolic	EPS	Rock wool	Glass wool	PUF Panel
-30 to -20	160	180	260	230	230	160
-20 to -15	130	140	210	190	190	130
-15 to -4	110	120	170	150	150	100
-4 to +2	90	100	150	130	130	90
+2 to +10	80	90	120	110	110	80
+10 to +16	80	90	120	110	110	80

Table 4-6: Floors (mm)

Temp (°C)	PUF/PIR	Phenolic	EPS	Rock wool	Glass wool	PUF Panel
-30 to -20	110	130	180	160	160	110
-20 to -15	100	110	150	140	140	100
-15 to -4	80	90	130	120	120	80
-4 to +2	80	90	120	110	110	80
+2 to +10	50	50	70	60	60	50
+10 to +16	30	40	50	50	50	30

Summary: Recommended Insulation Thickness at 40–47°C Ambient

Table 4-7 provides a consolidated summary of recommended insulation thicknesses at design ambient conditions of **40–47°C and 85–95% RH** — more demanding than IS 661: 2000 — for all major insulation materials used in cold storage construction.

Material	Min. Density (kg/m^3)	-30 to -20°C	-20 to -15°C	-15 to -4°C	-4 to +2°C	+2 to +10°C	+10 to +16°C	+16°C & above
Rigid Polyisocyanurate Foam	32	110	90	75	60	45	30	30
Rigid Polyurethane Foam	36	110	90	75	60	45	30	30
PU Panel — close construction	40±2	120	100	80	60	60	—	—
PU Panel — green field (open area)	40±2	150	120	100	80	60	—	—
Rigid Phenolic Foam	50	110	90	75	60	45	30	30
Expanded Polystyrene (EPS)	18	175	150	125	100	75	50	50
Bonded Rockwool	48	175	150	125	100	75	50	50

PUF and PIR offer the best insulation performance per unit thickness. EPS and Rockwool require significantly greater thickness — up to 175 mm at -30 to -20°C — to achieve equivalent thermal resistance. Open-area (green field) PU panels require 150 mm at the coldest range due to higher ambient exposure.

Insulation Thickness for Piping: Condensation Control

Refrigerant lines at temperatures below ambient must be insulated to minimise heat gain, control surface condensation, prevent ice accumulation, reduce noise, and ensure personal safety. Refrigeration systems typically operate from +5°C to as low as -45°C. Since moisture is a good thermal conductor, its presence is highly detrimental condensation can lead to complete system failure. In most refrigeration applications, the design thickness is therefore dictated by what is needed to **prevent condensation**, rather than by economic payback alone.

Vapour Pressure is the Driver

Just as temperature difference drives heat flow, **vapour pressure difference** drives moisture migration. Selection based on RH alone is incorrect selection must be based on vapour pressure difference, absolute humidity, or dew point temperature. Air at 40°C/20% RH contains nearly double the moisture (9.24 g/kg) of air at 5°C/95% RH (5.14 g/kg).

Dew Point is the Design Criterion

For selecting insulation thickness for piping, the surface temperature after insulation must be **above the dew point temperature** of the surrounding air. Table 4-10 (BS 5422: 1990) provides dew point temperatures at ambient conditions from -20°C to +50°C and RH from 50% to 95%, eliminating the need for repeated psychrometric chart lookups.

Installation Without Air Gaps

Correct installation is critical. If air — which always contains moisture in gaseous form — can reach the coolant pipe, condensation will still form. The vapour retarder must be **continuous and effective 100% of the time**. System operation is generally continuous, so vapour drive is unidirectional; water vapour that condenses remains in the insulation.

Piping Insulation: Thickness Calculation & Design Data

Condensation Prevention Formula

At steady state, heat flux from ambient air to the insulation surface equals heat flux from the insulation surface to the cold pipe:

$$t > \frac{k}{h} \frac{T_{dp} - T_{cold}}{T_{amb} - T_{dp}}$$

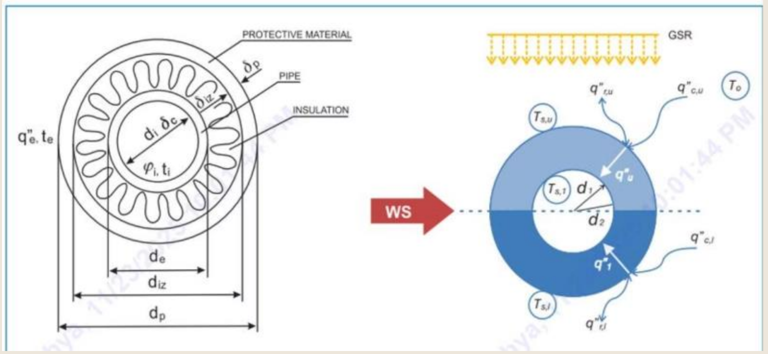
k = thermal conductivity of insulation (0.032 W/m·K at 23°C mean temp)
h = surface heat transfer coefficient (6.81 $W/m^2 \cdot K$)
T_amb = ambient dry bulb temperature
T_cold = fluid temperature inside pipe
T_dp = dew point temperature of ambient air

Lucknow example: 43.3°C DB, 22.8°C WB, dew point 12.4°C. Chilled water pipe at +5°C → min. thickness 1.53 mm. Liquid ammonia at -20°C → min. thickness 6.66 mm. Actual recommended thickness is higher, accounting for heat gain, rigidity, and environmental damage.

Table 4-13: Recommended PUF/PIR Thickness for Pipes at 38°C Ambient (mm)

Pipe Size (mm)	+5°C	-7°C	-20°C	-30°C	-40°C	-50°C
15	25	40	40	50	50	65
25	25	40	50	50	65	65
50	40	40	50	65	75	75
100	40	50	65	75	90	90
150	50	65	75	75	90	100
200	50	65	75	90	100	115
250	50	65	75	90	100	115
Flat Surface	50	75	90	100	120	140

Source: ASHRAE Refrigeration Handbook. Insulation is applied in multiple layers to accommodate expansion/contraction and because single-layer materials of sufficient thickness are often not manufactured.



Nitrile Rubber Insulation Thickness for Vessels & Pipes

Tables 4-14 through 4-17 give the insulation thickness for nitrile rubber insulation for vessels and pipes, both for standard inland conditions and for more demanding **coastal areas** (higher humidity requiring greater thickness).

Table 4-14: Nitrile Rubber — Vessels (Inland)

Vessel Type	Temp (°C)	Min. (mm)	Nominal (mm)
Water chiller	5	39	41
Brine flooded chillers	-10	66.4	70
Brine chillers	-20	84	89
Ammonia flooded L.P. vessels	-40	117.9	125
Inter-stage vertical coolers	-15	75.3	77

Table 4-16: Nitrile Rubber — Vessels (Coastal)

Vessel Type	Temp (°C)	Min. (mm)	Nominal (mm)
Water chiller	5	48.4	50
Brine flooded chillers	-10	79.5	83
Brine chillers	-20	99.5	100
Ammonia flooded L.P. vessels	-40	137.7	138
Inter-stage vertical coolers	-15	90	100

Table 4-15: Nitrile Rubber — Pipes (Inland, selected temperatures)

Pipe (mm)	-40°C Min.	-40°C Nom.	+5°C Min.	+5°C Nom.
25	52.5	56	21.9	25
50	59.7	64	24.4	25
100	68.1	70	26.9	32
200	76.3	79	28.9	32
250	78.9	79	29.4	32

Table 4-17: Nitrile Rubber — Pipes (Coastal, selected temperatures)

Pipe (mm)	-40°C Min.	-40°C Nom.	+5°C Min.	+5°C Nom.
25	68.2	69	30.2	32
50	77.8	80	34	34
100	89.2	91	37.9	38
200	101	102	41.4	45
250	105	105	42.4	32

Coastal area installations require significantly greater nitrile rubber thickness — approximately 15–25 mm more — due to higher ambient humidity and vapour pressure driving condensation risk.

Equivalent Thickness for Different Insulation Materials

Different insulation materials can achieve the same insulation effect if their thickness is selected appropriately. The figures below compare the thicknesses required to achieve equivalent thermal performance. Polyurethane consistently requires the least thickness, while foam glass, phenolic, and cellular concrete require the most. Application of insulation on vessels and pipes is covered in Chapter 6, *Industrial Applications*.

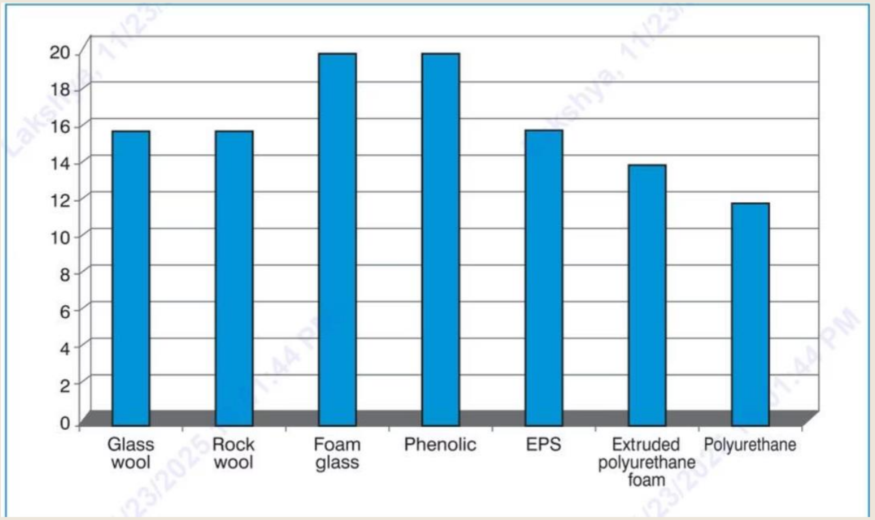


Figure 4-4: Relative thickness comparison. Polyurethane (12) requires the least; Foam glass and Phenolic (20) require the most to achieve equivalent insulation.

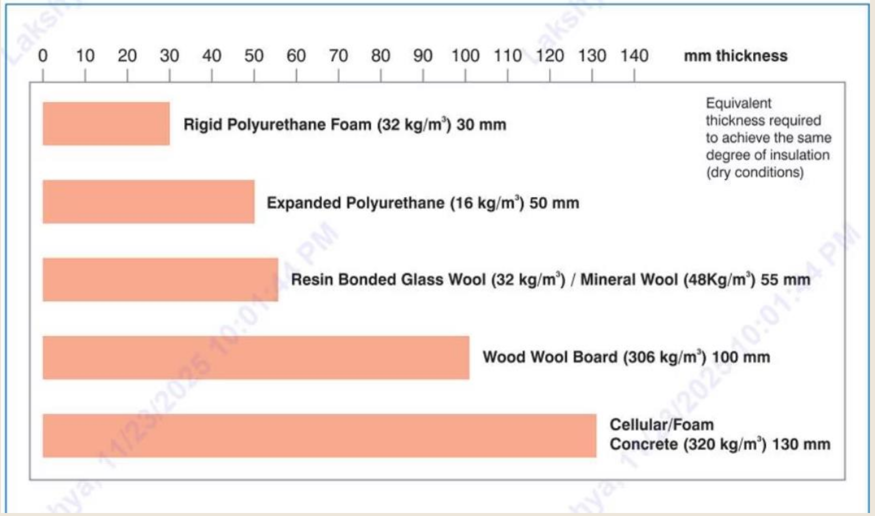


Figure 4-5: Equivalent thickness (dry conditions): Rigid PUF (32 kg/m³) = 30 mm; Expanded PU (16 kg/m³) = 50 mm; Glass/Mineral Wool = 55 mm; Wood Wool Board = 100 mm; Cellular Concrete = 130 mm.

30 mm

Rigid PUF

32 kg/m³ — most efficient insulation material per unit thickness

55 mm

Glass/Mineral Wool

32–48 kg/m³ — nearly double the thickness of PUF for same performance

100 mm

Wood Wool Board

306 kg/m³ — over 3× the thickness of rigid PUF required

130 mm

Cellular Concrete

320 kg/m³ — highest thickness requirement, over 4× that of rigid PUF

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